T2 Доп расчёты.

c) $\xi \sim p(x)$

$$p(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$M\xi = \int_0^{+\infty} x e^{-x} dx = \Gamma(2) = 1! = \boxed{1}$$

$$M\xi^2 = \int_0^{+\infty} x^2 e^{-x} dx = \Gamma(3) = 2! = \boxed{2}$$

$$\Downarrow$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2 - 1 = \boxed{1}$$

По простейшей УПТ получаем: ↓ см. далее

$$\frac{\sum_{i=1}^{n}\xi_i - \mathcal{M}\left(\sum_{i=1}^{n}\xi_i\right)}{\sqrt{\mathcal{D}\left(\sum_{i=1}^{n}\xi_i\right)}} \rightsquigarrow \eta \sim N(0,1)$$

$$\frac{\frac{1}{n}\sum_{i=1}^{n}\xi_i - \frac{1}{n}\mathcal{M}\left(\sum_{i=1}^{n}\xi_i\right)}{\sqrt{\frac{\mathcal{D}\left(\sum_{i=1}^{n}\xi_i\right)}{n^2}}} = \frac{\frac{1}{n}\sum_{i=1}^{n}\xi_i - \frac{n}{n}\mathcal{M}(\xi_1)}{\sqrt{\frac{n\,\mathcal{D}(\xi_1)}{n^2}}} =$$

$$= \frac{\frac{1}{n}\sum_{i=1}^{n}\xi_i - \mathcal{M}(\xi_1)}{\sqrt{\frac{\mathcal{D}(\xi_1)}{n}}} = \frac{\frac{1}{n}\sum_{i=1}^{n}\xi_i - 1}{\frac{1}{\sqrt{n}}} \rightsquigarrow \eta \sim N(0,1)$$

$n = 25$

$$\frac{1}{n}\sum_{i=1}^{n}\xi_i \rightsquigarrow \frac{1}{\sqrt{n}}\eta + 1 \rightsquigarrow N\left(1, \frac{1}{n}\right) = N\left(1, \frac{1}{25}\right)$$

е) плотность распределения медианы.

т.е $\xi_{(13)} \sim \sum_{i=13}^{25} C_{25}^{i} (F(x))^{i} \cdot (1-F(x))^{25-i} = \Phi(x)$

$$\tilde{p}(x) = \Phi'(x) = \sum_{i=13}^{25} C_{25}^{i} \cdot \Big(i \cdot (F(x))^{i-1} p(x) (1-F(x))^{25-i} -$$

$$- (25-i) \cdot (F(x))^{i} \cdot (1-F(x))^{24-i} \cdot p(x) \Big) =$$

$$= \sum_{i=13}^{25} C_{25}^{i} \cdot F(x)^{i-1} (1-F(x))^{24-i} \cdot p(x) \Big(i \cdot (1-F(x)) -$$

$$- (25-i) \cdot F(x) \Big) =$$

$$= p(x) \cdot \sum_{i=13}^{25} C_{25}^{i} \cdot F(x)^{i-1} (1-F(x))^{24-i} (i - 25F(x))$$